

Machine Learning Predictive Model

Customer Demand
Forecasting



ID Juicer is a health-focused retail company specializing in the sale of **sugar-free, fresh-from-the-source beverages**. We offer a diverse selection of premium healthy drinks, including cold-pressed juices, natural coconut water, and artisanal nut milk. Our mission is to seamlessly provide consumers with delicious, convenient products that support and promote a **healthy lifestyle**.



The defining operational constraint of ID Juicer is the **extreme perishability** of its products, limited to a **critical 5-to-7-day shelf life**. This inherent characteristic dictates that the entire supply chain and inventory management system must operate with **uncompromising speed and precision**.



The company must constantly solve a critical production puzzle:

- **Overproduction** directly results in costly spoilage and unavoidable **inventory waste**, severely eroding profit margins.
- Conversely, **Underproduction** to mitigate waste guarantees frequent **stockouts**, leading to lost sales revenue and diminishing customer satisfaction.

This tight operational window demands sophisticated, real-time demand forecasting and flawlessly executed cold chain logistics to perpetually balance the risk of **waste against the certainty of lost opportunity**.



In this project, I will try to solve this problem by **forecasting the customer demand of each ID Juicer product**. By knowing the demand, the company can achieve optimal inventory levels to eliminate both costly spoilage (waste) and revenue-losing stockouts (shortages), thereby maximizing profitability.



Model Development

[23]:

	Date	Day_of_Week	is_weekend	pay_day	product_category	product_name	unit_cost	discount_rate	Selling_Price	Total_Units_Sold	Total_Revenue	Profit
0	2025-01-01	Wednesday	0	1	Freshy Coconut	Coconut Water	8000	0.0	18000.0	77	1386000.0	770000.0
1	2025-01-01	Wednesday	0	1	Freshy Coconut	Kelapa Ijo	9000	0.0	20000.0	56	1120000.0	616000.0
2	2025-01-01	Wednesday	0	1	Nut Milk	Choco Blast	19000	0.1	40500.0	44	1782000.0	946000.0
3	2025-01-01	Wednesday	0	1	Nut Milk	Vanilla Shake	19000	0.2	36000.0	29	1044000.0	493000.0
4	2025-01-01	Wednesday	0	1	Juicy Juice	Tropical Vibes	17000	0.0	40000.0	32	1280000.0	736000.0
...	...	...	...	...	...	...	...	...	...	...	...	...
2333	2025-11-30	Sunday	1	1	Nut Milk	Choco Blast	19000	0.0	45000.0	58	2610000.0	1508000.0
2334	2025-11-30	Sunday	1	1	Nut Milk	Vanilla Shake	19000	0.0	45000.0	26	1170000.0	676000.0
2335	2025-11-30	Sunday	1	1	Juicy Juice	Tropical Vibes	17000	0.1	36000.0	48	1728000.0	912000.0
2336	2025-11-30	Sunday	1	1	Juicy Juice	Dynamic Green	17000	0.1	36000.0	26	936000.0	494000.0
2337	2025-11-30	Sunday	1	1	Juicy Juice	Sun Kissed	17000	0.0	40000.0	44	1760000.0	1012000.0

2338 rows × 12 columns

The analysis will be based on a dataset containing all **daily unit sales (demand)** recorded from **January 2025 to November 2025**. This data includes various features critical for determining demand fluctuations, such as **pay_day**, **is_weekend**, and **discount_rate**. These factors will be used to build a robust model capable of predicting future sales with high accuracy.

Model Development

```
[9]: df_predict.head()
```

```
[9]:
```

	Day_of_Week	is_weekend	pay_day	product_category	product_name	unit_cost	discount_rate	Selling_Price	Total_Units_Sold	Month	Day_of_Year
0	Wednesday	0	1	Freshy Coconut	Coconut Water	8000	0.0	18000.0	77	1	1
1	Wednesday	0	1	Freshy Coconut	Kelapa Ijo	9000	0.0	20000.0	56	1	1
2	Wednesday	0	1	Nut Milk	Choco Blast	19000	0.1	40500.0	44	1	1
3	Wednesday	0	1	Nut Milk	Vanilla Shake	19000	0.2	36000.0	29	1	1
4	Wednesday	0	1	Juicy Juice	Tropical Vibes	17000	0.0	40000.0	32	1	1

The initial dataset has been **transformed and refined** for model training. Key steps included:

- **Feature Engineering:** Defining new time-based features, specifically the **Month Number** and **Day of Year**, which are crucial for capturing seasonality and trend components in the demand data.
- **Feature Selection:** Dropping extraneous variables, such as **Total Revenue** and **Profit**, that are not directly used to predict demand and could potentially complicate the model.

This process has resulted in a **perfectly clean and finalized dataset**, which is now **ready for the split into training and testing sets** and subsequent model development.

Model Development

```
[12]: # Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

[13]: # Define the parameter distribution
param_dist = {
    'n_estimators': randint(100, 1000),
    'max_depth': [10, 20, 30, None],
    'min_samples_split': randint(2, 11),
    'min_samples_leaf': randint(1, 11),
    'bootstrap': [True, False],
    'max_features': [1.0, 'sqrt', 'log2']
}

[14]: # Initialize Random Forest model
rf = RandomForestRegressor(random_state=42, n_jobs=-1)

[15]: # Initialize Randomized Search CV
random_search = RandomizedSearchCV(
    estimator=rf,
    param_distributions=param_dist,
    n_iter=20,
    cv=5,
    scoring='neg_mean_absolute_error', # Optimize for minimizing MAE
    verbose=2,
    random_state=42,
    n_jobs=-1
)
```

First, i will divide our clean data into two parts, Training and Test Data

I'm using the **Random Forest Regressor**. This model is excellent because it can find complex connections between our features (like Pay Day and Discount) and the actual demand.

Tuning: A model has internal settings (parameters). Using **Randomized Search** to test many different settings quickly. This process makes sure to find the *best possible version* of the model, giving the most accurate forecast.

Model Development

	Metric	Value	Unit
0	R-squared (R2)	0.951318	(none)
1	Mean Absolute Error (MAE)	4.242040	units
2	Root Mean Square Error (RMSE)	7.184553	units
3	Mean Absolute Percentage Error (MAPE)	9.649218	%

The model's high **R2 of 0.95** and low **MAPE of 9.64%** demonstrate its strong **capability to accurately predict daily demand**, which is essential for solving the perishability problem. **The typical error is very small, averaging only 4.24 units (MAE).** This highly manageable margin for production planning allows ID Juicer to confidently balance the risks of waste and stockouts. While the **RMSE of 7.184 units indicates the model occasionally has larger prediction errors (likely during peak demand)**, the overall high accuracy ensures the forecast system is a reliable basis for making critical, time-sensitive production decisions.

Model Development

```
best_params = {'bootstrap': True, 'max_depth': 20, 'max_features': 1.0, 'min_samples_leaf': 2, 'min_samples_split': 6, 'n_estimators': 891}

best_rf = RandomForestRegressor(random_state=42, n_jobs=-1, **best_params)
best_rf.fit(X_train, y_train)
```

RandomForestRegressor

RandomForestRegressor(max_depth=20, min_samples_leaf=2, min_samples_split=6, n_estimators=891, n_jobs=-1, random_state=42)

```
import joblib
# Save the trained model object
joblib.dump(best_rf, 'demand_model.joblib')

# Save the list of feature columns (CRUCIAL for consistent feature ordering in deployment)
feature_columns = X.columns.tolist()
joblib.dump(feature_columns, 'model_features.joblib')
```

The last step is to prepare the best, most accurate model so ID Juicer can start using it to predict demand every day.

Saving the Final Model

- **Final Training:** I will train the model one last time using the **best settings** we found. This makes sure the final model is as smart as possible.
- **Saving the Model:** The finished model will be saved into a special file using the **joblib** tool. This makes it easy for the company's system to quickly load and use the model to make new predictions without having to train it again.
- **Saving the Features:** I will also save the list of all the important features used. This is a simple but important step that ensures the system feeds the right information to the model every time it needs a forecast, preventing errors.

This process turns our project into a **ready-to-use tool** that can immediately help ID Juicer manage production better.

Model Development

IDJuicer Demand Forecast Predictor (Dash)

Enter the parameters below and click 'Predict' to get the forecast.

Date & Operational Parameters

Prediction Date:

Is Weekend:

Is Pay Day:

Product & Pricing

Product Category:

Product Name:

Unit Cost (IDR):

Selling Price (IDR):

Discount Rate:

Get Demand Forecast

Click the 'Get Demand Forecast' button above.

The final step for practical implementation is integrating the trained model into a user-friendly application.

- The predictive model will be deployed via an **API (Application Programming Interface)** built using **Dash**.
- By using Dash, the complex machine learning model is transformed into an **intuitive operational dashboard**, ensuring that the high accuracy gained from the forecast translates seamlessly into **reduced waste and optimized inventory** in the real world.

Model Development

IDJuicer Demand Forecast Predictor (Dash)

Enter the parameters below and click 'Predict' to get the forecast.

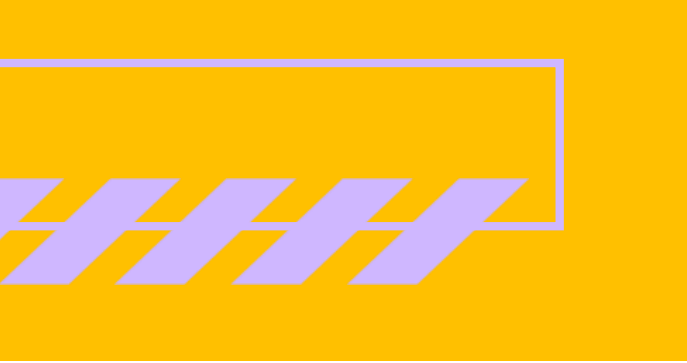
Date & Operational Parameters	Product & Pricing
Prediction Date: 2025-12-28	Product Category: Freshy Coconut
Is Weekend: Yes (1)	Product Name: Coconut Water
Is Pay Day: Yes (1)	Unit Cost (IDR): 8000
	Selling Price (IDR): 18000
	Discount Rate: <input type="range" value="0.05"/>

Predicted Demand for Coconut Water:

124 Units

Model validated with an average error (MAE) of 3.63 units.

This example shows this model ability to look into the future. For the coconut water stock on **Sunday, December 28, 2025**, after thoroughly analyzed all input features. The final chapter of model forecast predicts a robust sales day of **124 units**.



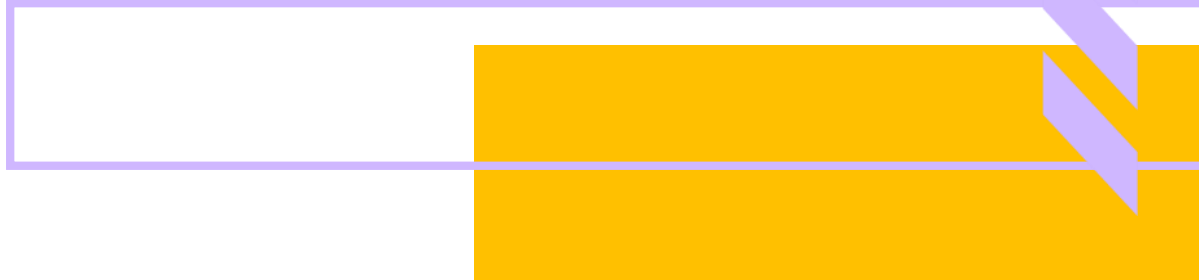
Conclusion

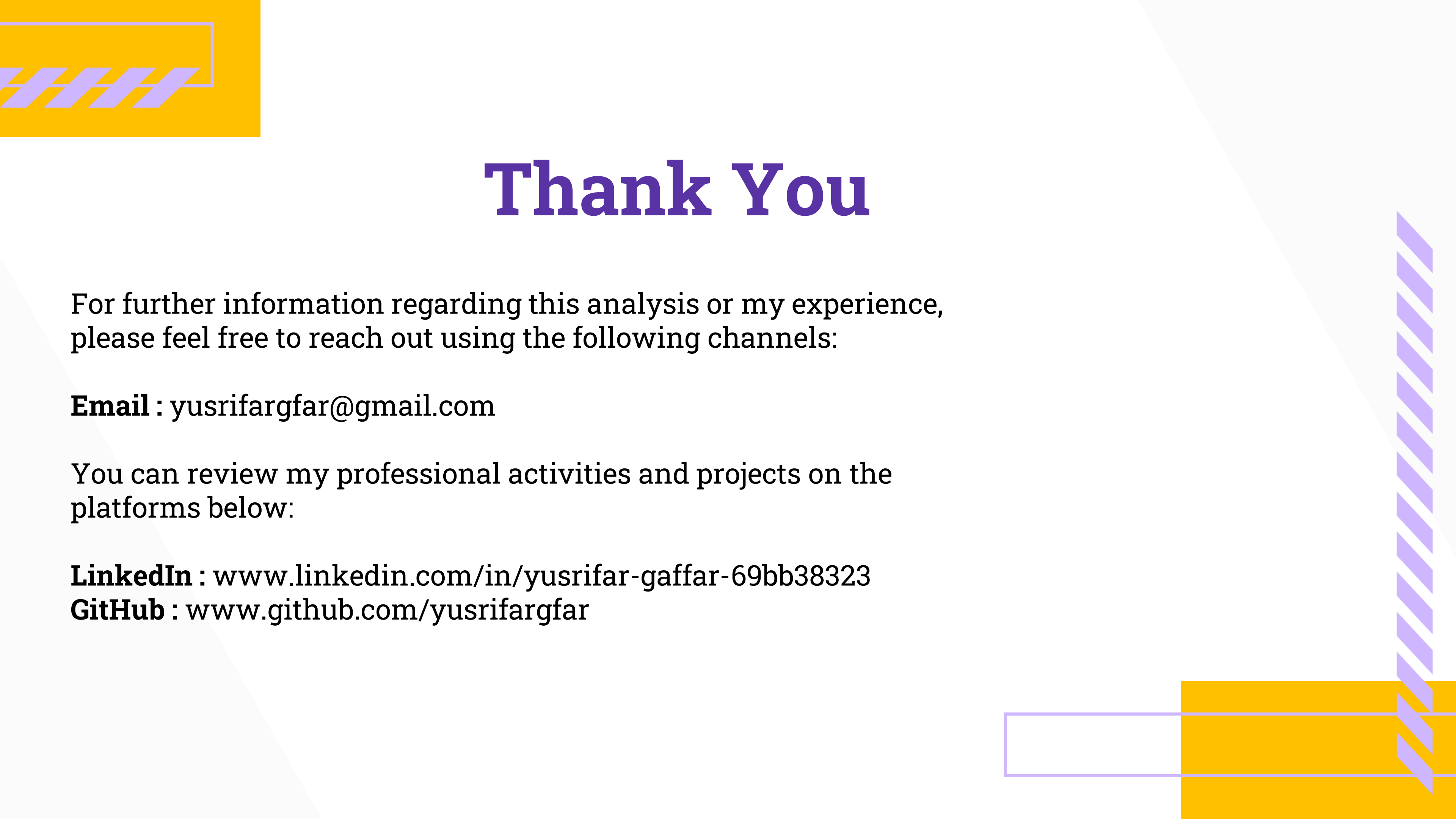
Based on how the model works.

Conclusion: Confidence in Prediction

- The example demonstrates the model's capacity to deliver a specific and quantifiable **demand forecast** for a future date, specifically predicting sales of **124 units** of coconut water for Sunday, December 28, 2025. This result is not a guess, but a data-driven projection based on systematically inputting all required market features.

Strategic Suggestions for the Company

- The forecast of **124 units** gives the company three key areas for action:
 - **Optimize Inventory:** Use the number 124 to **stock exactly what is needed**. This avoids revenue loss from stockouts while reducing waste and storage costs from overstocking.
 - **Streamline Operations: Adjust staffing and resources** (like delivery trucks or warehouse shifts) to match the predicted sales volume for December 28th, improving efficiency and service quality.
 - **Validate the Model:** After the day passes, **track and compare** the actual sales to the 124 unit prediction. This crucial step confirms the model's accuracy and provides data needed for continuous improvement.
- 



Thank You

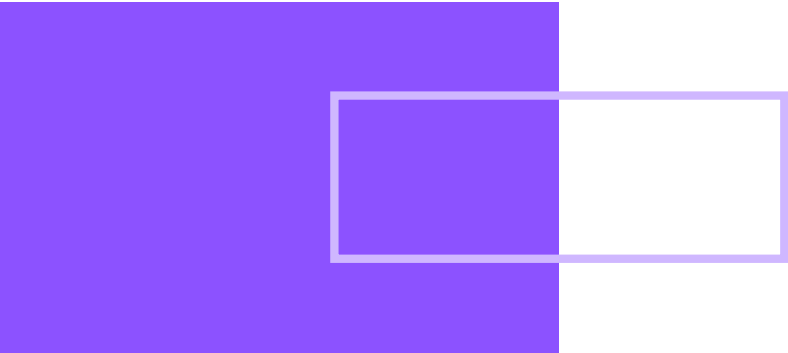
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